The Bystander Effect in the US–China Trade War: An Internal Meta-Analysis of Global Export Reallocations

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April 2026

1 Introduction

Between 2018 and 2019, the United States and China imposed successive waves of tariff increases on each other's exports, covering roughly $450 billion in bilateral trade flows and reversing a multi-decade trend toward lower trade barriers [Amiti et al., 2019]. A central empirical question raised by this episode is the extent to which third-party (bystander) countries absorbed the trade displaced by US–China tariffs and hence reshaped global patterns of specialization. If bystanders are close substitutes for Chinese varieties in the US market, and for US varieties in the Chinese market, the trade war should have generated positive reduced-form elasticities of bystander exports with respect to the bilateral tariff shocks. The sign, size, and distribution of these elasticities across countries and sectors is informative not only about the short-run consequences of the trade war, but also about the shape of global supply curves and the structure of product level substitution in modern supply chains.

This paper replicates and extends the empirical analysis of Fajgelbaum et al. [2023] (hereafter FGKKT). The authors estimate reduced-form elasticities of bystander exports to the United States, to China, and to the rest of the world (RW) with respect to the four tariff shocks of the trade war, using bilateral HS6-product-level trade data for the top 50 non-oil exporters over 2014–2019. Their central finding is that, on average, bystanders increased global exports in products targeted by higher US–China tariffs: the elasticity of bystander exports to RW with respect to the US tariff on China is approximately 0.20 (standard error 0.08), and with respect to the Chinese tariff on the US is approximately 0.29 (standard error 0.08). FGKKT interpret this pattern as evidence that the trade war created net trade opportunities rather than simply redistributing fixed volumes, and they show that the country specific component of these elasticities explains roughly 80% of the cross-country variation in export growth in targeted products.

The focal metric of my internal analysis is the common, pooled tariff elasticity of bystander exports to the rest of the world in response to the US tariff on China, which I denote β. This is the central parameter of Panel C of FGKKT's Figure 1 and the reduced-form counterpart of the coefficient $\beta^{RW}_{1i\omega}$ in their heterogeneous specification (their equation (10)). It directly measures the degree to which bystanders substituted for Chinese exports in global markets when the US

imposed tariffs on China. Because the authors emphasize average responses and country-level heterogeneity, but do not separately report how β varies by broad sector or by geographic region, the robustness of their central conclusion to these two dimensions of heterogeneity is an open empirical question.

My contribution is twofold. First, I successfully replicate the baseline point estimates of FGKKT's Figures 1 and 4 using the authors' public replication archive. [1] I was unable to reproduce Figures 2 and 3 due to hardware constraints, and it does not affect the baseline point estimates used below. Second, I design and estimate an internal meta-analysis of the bystander elasticity along two dimensions: sensitivity to modeling choices (dropping covariates, broadening fixed effects, and splitting the sample into agriculture versus non-agriculture) and regional heterogeneity across Asian, European, and American subsamples.

Three findings emerge. The pooled baseline estimate, $\hat{\beta} = 0.145$ (standard error 0.089), lies within the sampling range of the FGKKT headline of 0.20 and moves only modestly when covariates are dropped ($\hat{\beta} = 0.149$) or when fixed effects are broadened to the country level ($\hat{\beta} = 0.168$); the pooled specification is not fragile to these choices. The sectoral decomposition, however, reveals a sharp asymmetry: the estimated elasticity in agriculture alone is $\hat{\beta} = 0.360$, more than twice the pooled estimate and more than three times the non-agricultural estimate of $\hat{\beta} = 0.110$. This is consistent with the prominence of Chinese retaliatory tariffs on US agricultural products such as soybeans, pork, and dairy, which created concentrated opportunities for third-country agricultural exporters to displace US sales to China.

The regional decomposition reveals an even starker pattern: the Asian subsample elasticity is $\hat{\beta} = 0.434$, roughly three times the pooled baseline; the European estimate of $\hat{\beta} = 0.092$ is statistically indistinguishable from zero; and the American estimate of $\hat{\beta} = -0.223$ is actually *negative*, indicating that American bystanders' exports to the rest of the world *contracted* in products subject to higher US tariffs on China. The geographic concentration in Asia is consistent with the centrality of East and Southeast Asian economies in the supply chains linking intermediate inputs to China and final goods to the US market [Flaaen et al., 2020]. The negative American response, although noisily estimated, is consistent with upward-sloping export supply curves among Latin American bystanders under which reallocation toward surging US and Chinese demand pulls exports away from third-country markets. Taken together, these results qualify the headline narrative of FGKKT: the statement that "bystanders on average gained" is correct, but the average is essentially an Asian-agricultural average, with the European response muted and the American response moving in the opposite direction.

The remainder of the paper is organized as follows. Section 2 describes the data and variable construction used by FGKKT and retained in my analysis. Section 3 formalizes the authors' reduced-form specification and identifies the parameter of interest. Section 4 describes my internal meta-analysis design and the meta-regression equation used to summarize sensitivity. Section 5 presents the results, including one combined table of estimates and two forest plots. Section 6 concludes.

[1] See the project's GitHub repository for the full replication/analysis files, including the scripts used to generate Figures 1 and 4.

2 Data

The empirical analysis uses the bilateral HS6 product-level trade dataset assembled by Fajgelbaum et al. [2023], which combines UN Comtrade trade flows with a harmonized set of trade-war-era tariff changes.

Trade data. The underlying trade flows are drawn from UN Comtrade for the years 2014–2019. Trade is measured at the HS6 product level (ω), which yields 5,203 distinct product codes after harmonization across HS vintages using the standard UN concordances. To avoid contamination by commodity price cycles, the authors restrict the sample to the top 50 exporting countries excluding oil exporters; this trimmed sample accounts for approximately 95.9% of global trade and 70.5% of non-US, non-China trade. After dropping the US and China themselves from the bystander sample, 48 bystander exporters remain.

Tariff data. FGKKT compile four sets of tariff changes at the HS6 level, each scaled in proportion to its duration within the 24-month aggregation window: (i) *US tariffs on China*, $T^{US}_{CH,\omega}$, covering the Section 301, Section 232, and Section 201 increases imposed by the US in 2018–2019; (ii) *Chinese retaliatory tariffs on the US*, $T^{CH}_{US,\omega}$; (iii) *US tariffs on third countries*, $T^{US}_{i,\omega}$, including the steel and aluminum tariffs imposed on several non-Chinese partners; and (iv) *Chinese most-favored-nation (MFN) tariffs on third countries*, $T^{CH}_{i,\omega}$, which were cut during the same period and which Bown et al. [2019] argue were a strategic response to the US–China conflict. The first three series are taken from Fajgelbaum et al. [2020] and extended through the end of 2019; the fourth is constructed from Bown et al. [2019]. Tariff changes are expressed in log-change form, so that $\Delta \ln T^{US}_{CH,\omega}$ is the duration-weighted change in $\ln(1+\tau)$ across the relevant 24-month interval.

Sectoral classification. HS6 products are mapped to nine broad sectors—*agriculture*, apparel, chemicals, materials, machinery, metals, minerals, transport, and miscellaneous—using the sectoral crosswalk distributed with the replication archive. This classification is used both for constructing country-by-sector fixed effects in the main specification and for the sectoral subsamples examined in my meta-analysis.

Estimation sample and variable construction. The estimation file used in my regressions is `rf.dta`, one of the pre-processed datasets shipped in the authors' replication archive. The unit of observation is an exporter-HS6 pair. The dependent variable for the bystander-RW analysis, $\Delta \ln X^{RW}_{i\omega}$ (Stata name `dlv_rw`), is the log change in exporter i's exports of product ω to the rest of the world (all destinations other than the US and China) between the pre- and post-trade-war windows. The focal right-hand-side regressor, $\Delta \ln T^{US}_{CH,\omega}$ (`dlz_usch`), is the corresponding tariff change described above. The specification controls used below—lagged log export growth in the pre-trade-war window (`dlv_rw_lag`) and the variety-level size control (`mv_v_rw`)—are constructed exactly as described in the authors' online appendix. Detailed summary statistics for all variables are provided in Appendix A of Fajgelbaum et al. [2023].

3 Model Specification

This section describes the reduced-form specification of Fajgelbaum et al. [2023] and identifies the single coefficient that my internal meta-analysis targets.

3.1 The authors' pooled specification

FGKKT's pooled binscatter specification (their equation (8)) takes the form

$$\Delta \ln X_{i\omega}^{n} \;=\; \alpha \;+\; \beta^{n}\,\Delta \ln T_{\omega} \;+\; \varepsilon_{i\omega}^{n}, \tag{1}$$

where i indexes the 48 bystander exporters, ω indexes the 5,203 HS6 products, and $n \in \{US, CH, RW\}$ indexes the three destinations (United States, China, and rest of the world). The left-hand-side variable $\Delta \ln X_{i\omega}^{n}$ is the log change in exporter i's exports of product ω to destination n between the 2016/17 and 2018/19 windows. The right-hand-side variable $\Delta \ln T_{\omega}$ is the relevant tariff shock—either $\Delta \ln T_{CH,\omega}^{US}$ (the US tariff on China) or $\Delta \ln T_{US,\omega}^{CH}$ (the Chinese retaliation on the US), depending on the panel of Figure 1. Equation (1) imposes a common elasticity β^{n} across all bystander exporters and sectors. The Panel C estimate—bystander exports to the rest of the world as a function of the US tariff on China—yields $\hat{\beta}^{RW} = 0.20$ with a standard error of 0.08 in the authors' reported version, and is the headline average response I re-examine below.

3.2 The authors' heterogeneous specification

The full heterogeneous specification, from which the pooled version can be viewed as a restricted special case, is equation (10) of FGKKT:

$$\begin{aligned}\Delta \ln X_{i\omega}^{n} = {} & \beta_{1i\omega}^{n}\,\Delta \ln T_{CH,\omega}^{US} + \beta_{2i\omega}^{n}\,\Delta \ln T_{US,\omega}^{CH} + \beta_{3i\omega}^{n}\,\Delta \ln T_{i,\omega}^{US} + \beta_{4i\omega}^{n}\,\Delta \ln T_{i,\omega}^{CH} \\ & + \alpha_{ij(\omega)}^{n} + \Omega^{n}\mathrm{SIZE}_{i\omega} + \pi^{n}\,\Delta \ln X_{i\omega,t-1}^{n} + \varepsilon_{i\omega}^{n},\end{aligned} \tag{2}$$

where $j(\omega)$ denotes the broad sector containing product ω, $\alpha_{ij(\omega)}^{n}$ are origin-destination-sector fixed effects, $\mathrm{SIZE}_{i\omega}$ is a vector of pre-war trade-flow size controls, and $\Delta \ln X_{i\omega,t-1}^{n}$ is lagged export growth from the 2014/15 $\rightarrow$ 2016/17 placebo window. The four coefficients $\beta_{zi\omega}^{n}$ ($z = 1, \ldots, 4$) are allowed to vary by exporter and sector through the linear decomposition $\beta_{ziw}^{n} = \beta_{zi}^{n} + \beta_{zj(\omega)}^{n} + \Gamma_{z}^{n}\mathrm{SIZE}_{zi\omega}^{n}$ imposed in the authors' equation (9). The identifying assumption is that, conditional on origin-destination-sector fixed effects, size controls, and lagged export growth, potential export growth across products would have evolved in parallel in the absence of the tariff shocks. FGKKT defend this assumption by showing that pre-war export growth is broadly uncorrelated with the subsequent tariff changes.

3.3 The specification estimated in this paper

My internal meta-analysis targets the reduced-form, pooled-coefficient analog of equation (2) for the rest-of-the-world destination. Concretely, I estimate

$$\Delta \ln X_{i\omega}^{RW} \;=\; \beta\,\Delta \ln T_{CH,\omega}^{US} \;+\; \pi\,\Delta \ln X_{i\omega,t-1}^{RW} \;+\; \Omega\,\mathrm{SIZE}_{i\omega}^{RW} \;+\; \alpha_{ij(\omega)} \;+\; \varepsilon_{i\omega}, \tag{3}$$

with observations weighted equally and standard errors clustered at the HS6 product level. Equation (3) differs from equation (2) in two respects. First, it restricts the coefficient on the US tariff on China to be common across exporters and sectors, $\beta^{RW}_{1i\omega} \equiv \beta$; this is the pooled analog of the heterogeneous specification and directly corresponds to the slope in Panel C of FGKKT's Figure 1. Second, it omits the three remaining tariff shocks ($\Delta \ln T^{CH}_{US,\omega}$, $\Delta \ln T^{US}_{i,\omega}$, $\Delta \ln T^{CH}_{i,\omega}$), treating their effects as absorbed by the fixed effects and controls. This simplification is standard for the bin-scatter specification and is adopted to keep the meta-analysis focused on a single, interpretable coefficient. My baseline replication of equation (3) on the pooled sample of 48 bystanders yields $\hat{\beta} = 0.145$ with a clustered standard error of 0.089—an estimate that lies within the sampling range of FGKKT's headline 0.20 and that the five specifications of Section 4 perturb in economically interpretable ways.

4 Analysis Design

The central question of my internal analysis is whether the pooled bystander elasticity β in equation (3) conceals systematic heterogeneity along two dimensions that FGKKT do not separately report: sector (agriculture vs. non-agriculture) and geography (Asia vs. Europe vs. Americas). I investigate this question through two sets of specification perturbations and summarize the results using a simple meta-regression framework.

4.1 Sectoral and modeling sensitivity (Analysis 1)

The first block of specifications re-estimates equation (3) on the pooled bystander sample while varying one modeling choice at a time. Specification **(S1)** is the *baseline*: equation (3) on all 48 bystanders, with both controls and country-by-sector fixed effects. **(S2)** drops the lagged export growth and size controls, retaining only the fixed effects, to isolate their contribution. **(S3)** replaces country-by-sector fixed effects with exporter-country fixed effects, which absorbs less within-country heterogeneity and therefore relies more heavily on across-product tariff variation within each exporter. **(S4)** restricts the sample to HS6 products classified as *agriculture* by FGKKT's sectoral crosswalk, and **(S5)** restricts to its complement (all non-agricultural products, primarily manufacturing). Specifications (S1)–(S3) gauge sensitivity to standard modeling choices; (S4)–(S5) decompose the pooled estimate into its agricultural and non-agricultural components.

4.2 Regional heterogeneity (Analysis 2)

The second block of specifications partitions the 48 bystanders into three mutually exclusive geographic regions—Asia, Europe, and the Americas—and re-estimates the baseline equation (3) within each region. The regional assignment is based on ISO3 country codes and follows conventional continental definitions. Specifications **(S6)**, **(S7)**, and **(S8)** estimate equation (3) on the 14 Asian, 18 European, and 9 American bystanders, respectively.[2]

Geographic heterogeneity is of substantive interest because Asian economies, and especially East and Southeast Asian ones, are deeply integrated into supply chains in which Chinese assembly

[2] See `02_meta_analysis.do` and `03_regional_meta_analysis.do` in the `code/` folder for detailed analysis specifications.

plants import intermediate inputs from regional partners [Flaaen et al., 2020]. If the bystander response to US tariffs on China reflects the migration of processing activity out of China, one would expect the elasticity to be concentrated in the Asian subsample. By contrast, European and American economies trade with China through arm's-length channels with thinner intermediate-input linkages, and would be expected to exhibit smaller substitution responses.

4.3 Meta-regression summary

To summarize the eight resulting point estimates and their standard errors in a compact form, I treat the outcomes of specifications (S1)–(S8) as eight draws from a meta-regression. For each specification $s \in \{1, \ldots, 8\}$, let $\hat{\beta}_s$ denote the estimated tariff elasticity and $\hat{\sigma}_s$ its clustered standard error. I report the forest-plot summary

$$\hat{\beta}_s = \beta_0 + \sum_k \gamma_k D_s^k + u_s, \qquad u_s \sim \mathcal{N}(0, \hat{\sigma}_s^2), \tag{4}$$

where D_s^k are indicator variables identifying which specification choice distinguishes s from the baseline (e.g., $D_4^{\text{Ag}} = 1$ for the agriculture-only specification, $D_6^{\text{Asia}} = 1$ for the Asian subsample, and so on). The intercept β_0 captures the baseline estimate, and the γ_k coefficients capture the shifts in the point estimate induced by each modeling choice or subsample restriction, with residual variance calibrated to the within-specification clustered variance $\hat{\sigma}_s^2$. Because the eight specifications are applied to partially overlapping samples and cannot be jointly estimated with a single meta-regression, I report equation (4) descriptively rather than as a fitted model: each row of Table 1 below corresponds to one $\hat{\beta}_s$, and the two forest plots in Figures 1 and 2 display the resulting $\hat{\beta}_s \pm 1.96\hat{\sigma}_s$ intervals.

5 Analysis Results

Table 1 presents the eight point estimates and standard errors that summarize the internal meta-analysis. Figures 1 and 2 display the same estimates as forest plots with 95% confidence intervals.

5.1 Sectoral and modeling sensitivity

Panel A of Table 1 and Figure 1 report the results of specifications (S1)–(S5).

The pooled baseline (S1) yields $\hat{\beta} = 0.145$ with a clustered standard error of 0.089, implying a 95% confidence interval of approximately $[-0.03, 0.32]$ that comfortably contains FGKKT's headline Panel C estimate of 0.20. Dropping the lagged export growth and size controls (S2) barely moves the point estimate, to 0.149; this suggests that once country-by-sector fixed effects are included, the residual predictive content of the pre-war controls for post-war export growth is small. Broadening the fixed-effect structure from country-by-sector to country (S3) raises the estimate modestly to 0.168 with a tighter standard error of 0.076, consistent with the fact that this specification absorbs less within-exporter variation and therefore uses a larger share of the cross-sectional tariff variation. The pooled elasticity is thus not fragile to the two most natural modeling perturbations. The qualitative conclusion of FGKKT—that bystanders on average reallocated exports toward targeted products—survives both.

Table 1. Internal Meta-Analysis of the Bystander Export Elasticity

	Specification	$\hat{\beta}$	Cluster SE
Panel A. Sectoral and modeling sensitivity			
(S1)	Baseline	0.145	(0.089)
(S2)	No covariates	0.149	(0.087)
(S3)	Broader fixed effects (country)	0.168	(0.076)
(S4)	Agriculture only	0.360	(0.226)
(S5)	Non-agriculture / Manufacturing	0.110	(0.096)
Panel B. Regional heterogeneity			
(S6)	Asia subsample	0.434	(0.153)
(S7)	Europe subsample	0.092	(0.110)
(S8)	Americas subsample	−0.223	(0.206)

Notes: Each row reports the coefficient $\hat{\beta}$ on $\Delta \ln T^{US}_{CH,\omega}$ from a separate estimation of equation (3). All specifications use `reghdfe` in Stata with standard errors clustered at the HS6 product level. (S1) is the pooled baseline with country-by-sector fixed effects and both controls. (S2) drops the controls. (S3) replaces country-by-sector fixed effects with country fixed effects. (S4) and (S5) restrict the product sample to agriculture and non-agriculture respectively. (S6)–(S8) restrict the exporter sample to Asia, Europe, and the Americas respectively, using the country-list groupings described in Section 4.
Source: Author's calculations using `rf.dta` from the Fajgelbaum et al. [2023] replication archive. Stata code is available in the accompanying GitHub repository.

The two sectoral specifications, however, reveal that this average masks a sharp asymmetry. Restricting the sample to agriculture alone (S4) more than doubles the point estimate, to $\hat{\beta} = 0.360$, although the standard error widens to 0.226, so the confidence interval barely excludes zero at the 95% level. Restricting to non-agriculture (S5), which contains the bulk of the HS6 products, delivers a point estimate of $\hat{\beta} = 0.110$, below the pooled baseline and roughly one-third of the agricultural estimate. Economically, a one-percentage-point increase in the log US tariff on China is associated with a 0.36% increase in bystander agricultural exports to RW and only a 0.11% increase in bystander non-agricultural exports. This ordering is consistent with the well-documented structure of the trade war itself: Chinese retaliation against the United States was disproportionately targeted at agricultural products, most notably soybeans, for which the primary alternative suppliers are Brazil, Argentina, and a small number of other large agricultural exporters [Bown et al., 2019]. Agricultural markets are also relatively homogeneous at the HS6 level, so the reallocation of demand across suppliers is mechanically more elastic than in differentiated manufactured goods.

5.2 Regional heterogeneity

Panel B of Table 1 and Figure 2 report the results of specifications (S6)–(S8).

The regional decomposition reveals a striking concentration of the bystander response in Asia, together with an unexpected sign reversal in the Americas. The Asian subsample (S6) delivers

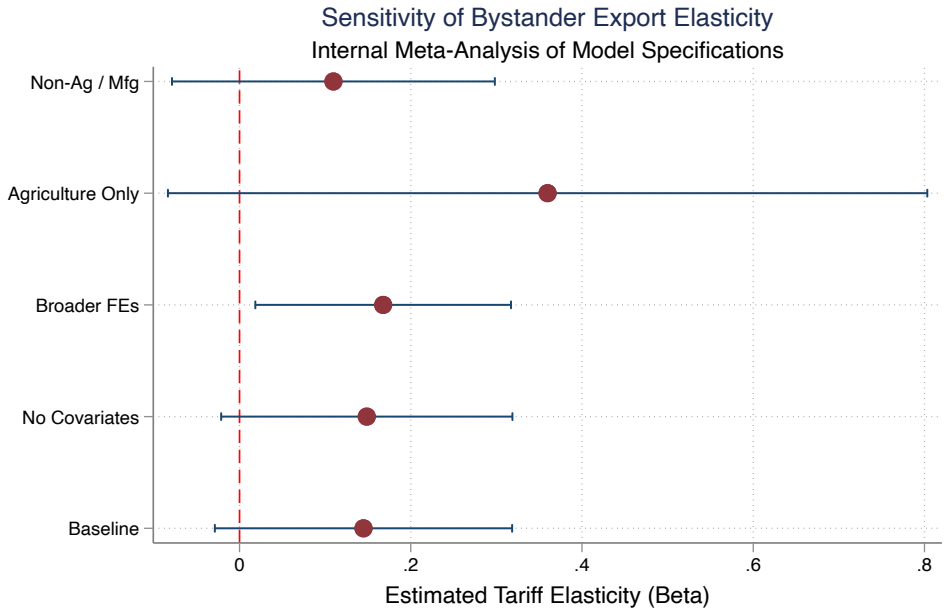

Figure 1. Sectoral and modeling sensitivity of the bystander export elasticity

Notes: Point estimates (markers) and 95% confidence intervals (whiskers) for the coefficient $\hat{\beta}$ on $\Delta \ln T^{US}_{CH,\omega}$ from specifications (S1)–(S5) of equation (3). The dashed vertical line marks zero. *Source:* Author's calculations using the Fajgelbaum et al. [2023] replication archive.

$\hat{\beta} = 0.434$ with a standard error of 0.153; the corresponding 95% confidence interval, $[0.13, 0.73]$, excludes zero at conventional levels and is centered on a value roughly three times the pooled baseline and more than twice the FGKKT headline of 0.20. The European estimate (S7) is $\hat{\beta} = 0.092$, close to the pooled baseline but with a standard error of 0.110 that leaves the confidence interval well centered on zero. The Americas estimate (S8) is $\hat{\beta} = -0.223$ with a standard error of 0.206: American bystander exports to the rest of the world actually *contracted* in products subject to higher US tariffs on China, and although the 95% confidence interval of $[-0.63, 0.18]$ does not exclude zero, the point estimate is an order of magnitude below the pooled baseline and opposite in sign.

The ordering Asia $>$ Europe $>$ Americas, with the latter turning negative, matches and sharpens the qualitative pattern of country-specific effects documented in FGKKT's Figure 4, where East and Southeast Asian economies such as Vietnam, Thailand, and Korea cluster in the substitution quadrants for both US and Chinese tariffs while several Latin American exporters lie in the complement quadrants. The Asian result confirms that the pooled average is disproportionately driven by a specific geographic subgroup: when Asia is excluded, the remaining bystanders as a group exhibit essentially no positive response to US tariffs on China.

More interestingly, the negative Americas point estimate, while noisy, is consistent with a mechanism embedded in FGKKT's own theoretical framework. Under upward-sloping export supply, an exporter that reallocates production toward surging demand in the US and Chinese markets must pull resources away from other destinations; the observed negative coefficient on exports to the rest of the world is exactly the sign predicted by that mechanism. More precisely, FGKKT themselves identify Ukraine and Colombia as countries whose exports complement Chinese and

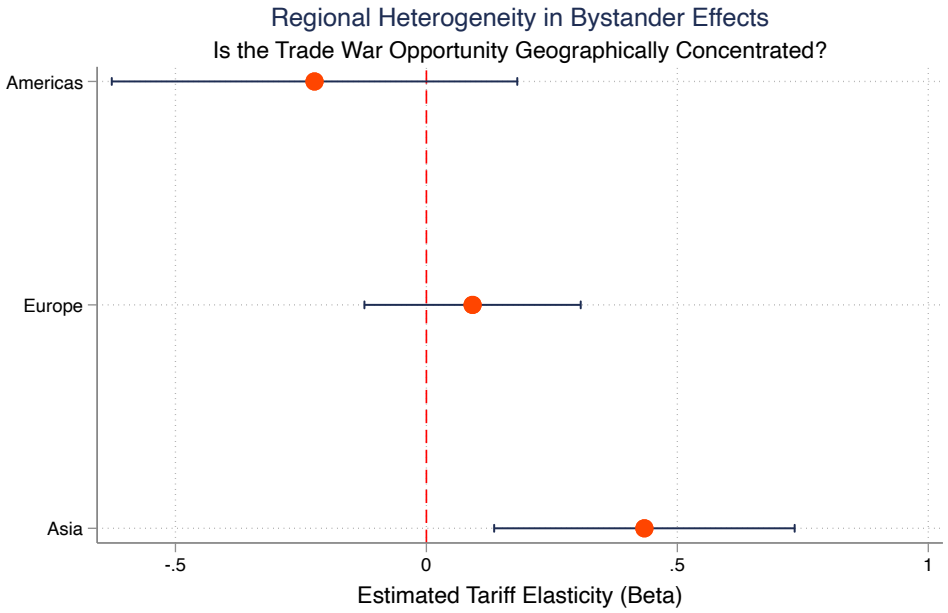

Figure 2. Regional heterogeneity in the bystander export elasticity

Notes: Point estimates (markers) and 95% confidence intervals (whiskers) for the coefficient $\hat{\beta}$ on $\Delta \ln T^{US}_{CH,\omega}$ from specifications (S6)–(S8) of equation (3), estimated separately on the Asian, European, and American bystander subsamples. The dashed vertical line marks zero. *Source:* Author's calculations using the Fajgelbaum et al. [2023] replication archive.

American varieties and operate along downward-sloping supply curves, a combination that the authors document as generating global export contractions in response to the trade war. The Americas subsample aggregates a set of economies whose product-level responses plausibly mix this mechanism with the upward-supply-and-substitution case, and the pooled negative estimate is consistent with either interpretation.

Panel A and Panel B cut the data in genuinely distinct ways. The major Asian bystanders—Vietnam, Malaysia, Thailand, and Korea—specialize in electronics, machinery, and apparel, none of which belong to the agricultural subsample. The agricultural response is largely driven by Latin American and Oceanic exporters such as Brazil, Argentina, and Australia, which are precisely the economies whose pooled elasticity turns negative when grouped into the Americas subsample. The two dimensions of heterogeneity are therefore complementary: the trade war's global reallocation benefited Asian manufacturing and global agriculture, but on net reduced exports from the Americas to the rest of the world.

6 Conclusions

The headline finding of Fajgelbaum et al. [2023]—that bystander countries, on average, increased their global exports in products targeted by US–China trade-war tariffs—rests on a pooled elasticity of approximately 0.20 with a standard error of 0.08. This paper has shown that this average, while real, conceals two economically meaningful dimensions of heterogeneity. Along the sec-

toral dimension, the elasticity is roughly three times larger in agriculture than in non-agriculture, consistent with the prominence of Chinese retaliatory tariffs on US agricultural exports and with the greater substitutability of agricultural products across suppliers. Along the geographic dimension, the pooled response masks a reversal of sign across regions: the Asian subsample elasticity of 0.434 is roughly three times the pooled baseline, the European estimate of 0.092 is close to the baseline but statistically indistinguishable from zero, and the American estimate of -0.223 is negative, indicating that American bystanders' exports to the rest of the world contracted in tariff-targeted products.

These results qualify the policy narrative often associated with the original paper in a specific way. The statement that "bystanders gained" is not false, but it is an average that obscures a pattern of asymmetric gains: Asian economies with deep supply-chain integration into China's export processing networks, together with countries holding comparative advantage in targeted agricultural products, were the primary beneficiaries of the trade war's reallocation of global trade flows. European bystanders were largely unaffected at the product-level margin studied here, and American bystanders' responses were, if anything, negative—plausibly reflecting reallocation of limited export capacity away from the rest of the world toward expanding US and Chinese demand, a mechanism that arises naturally in the upward-sloping supply case of FGKKT's theoretical framework.

Several directions for further research follow naturally. The large agricultural elasticity raises the question of whether the gains accrued primarily through quantity or price adjustments at the exporter-product level, a decomposition that the current specification does not identify. The sharp ordering of the regional elasticities invites a structural investigation of the role of supply-chain depth, geographic proximity to China, and the composition of intermediate-input trade in driving the bystander response; FGKKT's heterogeneous specification (their equation (2)) provides a natural starting point. Finally, a fuller two-dimensional (sector $\times$ region) analysis would reveal whether, for example, agricultural gains accrue to Latin American exporters while manufacturing gains accrue to Asian ones—an exercise that, given the moderate sample sizes within each cell, would likely require the heterogeneous-coefficient machinery of equation (2) and is deferred to future work.

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